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Mechanical Properties Of Solids

- **Deforming Force:-** A force which changes the size or shape of a body is called as deforming force.

- **Elasticity:-** The property of a body to regain its original configuration after removal of deforming force, upto its elastic limit.

→ Steel is more elastic than rubber.

- **Plasticity:-** No tendency to regain previous shape and the body get deformed permanently.
  e.g Putty and mud.

\# In particle life, no body is perfectly elastic or plastic.

Elastic Behaviour of Solids

- When a solid is deformed, the atoms are displaced from their equilibrium position causing a change in interatomic distances.
- When force is removed, the body regains its original shape and size.

→ Spring-ball model explains, elastic behaviour of solids.

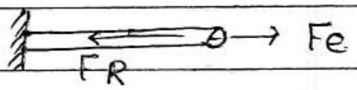
The stronger the interatomic Forces, the smaller will be the displacement of atoms From equilibrium positions and hence greater is elasticity of material.

Stress (σ)

→ The internal restoring Force per unit cross sectional area is called stress.

$$\text{Stress} = \frac{\text{Restoring Force}}{\text{Area}}$$

$$\boxed{\sigma = \frac{F_R}{A}}$$



$a = 0,\ F_R = F_e$.

- SI unit = Nm^{-2} = Pascal.
- Dimension :- $ML^{-1}T^{-2}$

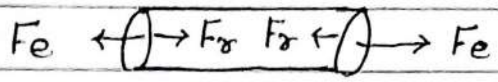
Types of Stress :-

1. Longitudinal stress

- The deForming Force will be acting along the length of the body.
- Longitudinal stress results in the change in the length of the body.

Two types of longitudinal stress :-

1. Tensile stress



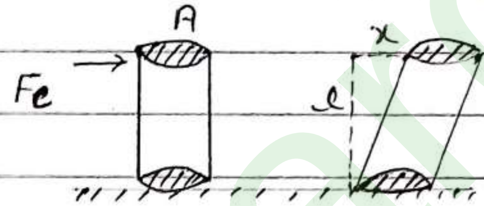
$$\sigma_T = \frac{F_R}{A} = \frac{F_e}{A}$$

2. Compressive

$$\sigma_c = \frac{F_R}{A}$$

2. Shearing / Tangential Stress

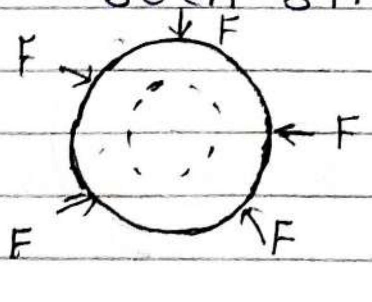
It arises from the shear force the component of the force vector parallel to the material cross section.



$$\sigma_s = \frac{F_t}{A}$$

3 Volumetric / Hydraulic Stress / Bulk stress

When the deforming force or applied force acts from all dimensions resulting in the change of volume of object then such stress is called volumetric stress.



$$\sigma_v = \frac{F}{A}$$

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Que-1 The ratio of radii of two wires of same material is 2:1. Find if they are stretched by the same force, the ratio of stress.

Sol^n

$$\sigma = \frac{F}{A} = \frac{F}{\pi r^2} \Rightarrow \sigma \propto \frac{1}{r^2}$$

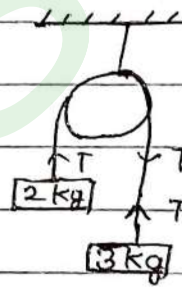
$$\frac{\sigma_1}{\sigma_2} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4} = 1:4$$

Q-2 The breaking stress of metal wire is $\frac{2.4 \times 10^3}{\pi}$ Nm^{-2}. The minimum radius of the wire used if it is not to brake. ($g = 10 m/s^2$)

Sol^n

$$\sigma \le \sigma_{breaking}$$

$$\frac{F}{A} \le \sigma_b$$



$$T = 2g\frac{m_1 m_2}{m_1 + m_2}$$

$$= \frac{2 \times 10 \times 2 \times 3}{5}$$

$$T = 24 N.$$

$$\frac{T}{\pi r^2} \le \sigma_b$$

$$\frac{\cancel{24}}{\cancel{\pi} r^2} \le \frac{\cancel{2.4}}{\cancel{\pi}} \times \frac{10^9}{10}$$

$$\frac{1}{r^2} \le 10^8 \Rightarrow \frac{r^2}{r} \le \frac{1}{10^8}$$

$$\boxed{r \le 10^{-4}}$$

Strain (ϵ)

The ratio of the change in any dimensions produced in the body to the original dimension is strain.

$$\left[\text{strain} = \frac{\text{change in configuration of object}}{\text{original configuration.}}\right]$$

- unit of strain = No unit (Dimensionless)

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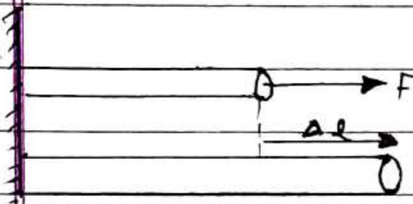
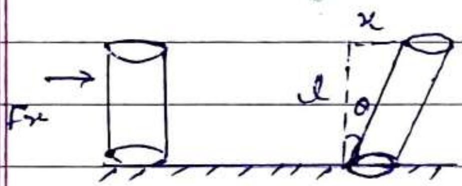
1. Longitudinal strain



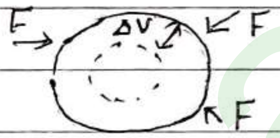
$$\epsilon = \frac{\Delta l}{l}$$

2. Shearing strain



$$\epsilon = \frac{x}{l} = \tan\theta$$

3. Volumetric / Bulk strain



$$\epsilon = \frac{\Delta V}{V}$$

Que-3 A wire of length 2.5 m has a percentage strain of 0.12% under a tensile force. The extension produced in the wire will be?

Solu. $\epsilon = \frac{\Delta l}{l} = \Delta l = \epsilon \times l$

$$= \frac{\cancel{0.012}^{\,3}}{\cancel{100}_{\,4}} \times \frac{\cancel{2.5}}{10} \times \frac{1}{1000} = 3 \times 10^{-4}$$

$$= 0.3 \times 10^{-3}$$

$$= 0.3\ mm.$$

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Q-4. If the angle of shear is 30° for a cubical body and change in length is $100\sqrt{3}$ cm. Find the volume of cubical body.

Soln Angle of sher, $\theta = 30°$

$\Delta l = 100\sqrt{3}$ cm

$V = ?$

$\tan\theta = \frac{\Delta l}{l}$

$\frac{1}{\sqrt{3}} = \frac{100\sqrt{3}}{l} \Rightarrow l = 300$ cm $= 3m.$

V. of cube $= a^3$
$= (3)^3$
$= 27\ m^3$

Q-5. A uniform cube is subjected to volume compression. If each side is decreased by 1% then bulk strain is?

Soln Side decreased by 1%.

$\Delta a = \frac{a}{100}$

$a' = a - \frac{a}{100}$ (New side)

Change in volume $= a^3 - a'^3$

$= a^3 - \left[a - \frac{a}{100}\right]^3$

$= a^3 - a^3\left[1 - \frac{1}{100}\right]^3$ $\left((1-x)^n = 1 - nx, \ x <<< 1 \right)$

$= a^3 - a^3\left[1 - \frac{3}{100}\right], = a^3\left[\cancel{1} - \cancel{1} + \frac{3}{100}\right]$

$\Delta V = \frac{3a^3}{100}$

$\epsilon = \frac{\Delta V}{V} = \frac{\frac{3\cancel{a^3}}{100}}{\cancel{a^3}} = \frac{3}{100} = 0.03$

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Que-6 A uniform sphere is compressed such that its radius decreased by 2% the volumetric strain (Bulk strain) is?

Solⁿ Radius is decreased by 2%.

$$\Delta r = \frac{2r}{100}$$

$$r' = r - \frac{2r}{100}$$

$$\Delta V = \frac{4}{3}\pi\left[r^3 - \left(r - \frac{2r}{100}\right)^3\right] = \frac{4}{3}\pi\left[r^3 - r^3\left[1 - \frac{2}{100}\right]^3\right]$$

$$= \frac{4}{3}\pi\left[r^3 - r^3\left[1 - \frac{6}{100}\right]\right]$$

$$= \frac{4}{3}\pi r^3\left[1 - 1 + \frac{6}{100}\right]$$

$$\varepsilon = \frac{\Delta V}{V} = \frac{\cancel{\frac{4}{3}\pi r^3} \times \frac{6}{100}}{\cancel{\frac{4}{3}\pi r^3}} = \frac{6}{100} = 0.06 \text{ Ans}$$

Elastic limit

→ The maximum stress within which the body completely regain its original size and shape after the removal of deforming force is is called Elastic limit.

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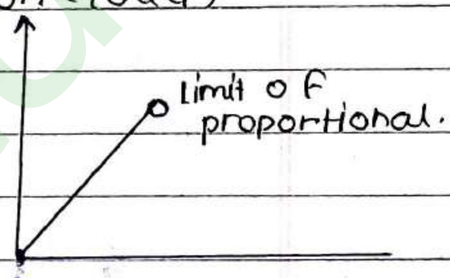
Hooke's Law

→ In 1679, Robert Hooke presented his law of elasticity which is now known as Hooke's law.

→ According to this law, For small deformation, stress in a body is directly proportional to strain within elastic limit.

Stress ∝ Strain (extension ∝ load)

Stress = E Strain



$$\left\{ E = \frac{\text{stress}}{\text{strain}} \right\}$$

E = Coefficeent or modulus of Elasticity
↳ Property of material

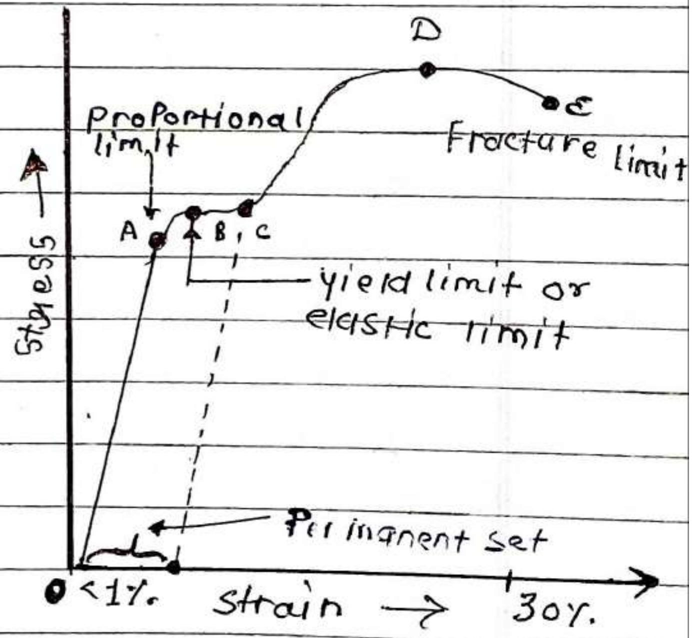
- Law is not valid for large values of strains.

Stress-Strain Curve

OA: stress ∝ strain upto A
A → Proportional limit.

AB: Stress ↑ Strain ↑
But proportionality ends

uptill point B



★ On removing deforming force, wire returns to original position, strain becomes zero if stress is reduced to zero.

B → Elastic limit or Yield point

After Elastic Limit

- Property of elasticity reduces.
- Tendency to return to original position decreases.

After B, on removal of deforming Force, body do not return to original position. (permant strain)

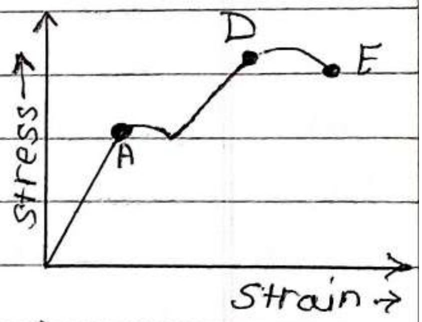
At point D

- Maximum stress / Ultimate tensile stress / (σ_U) braking stress.

At E → Fracture point.

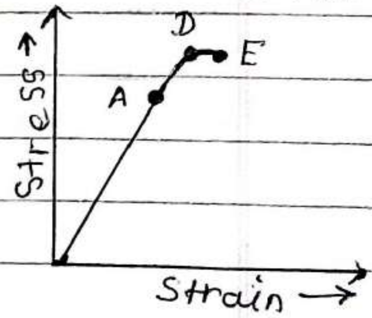
Different Nature of Stress- Strain curve

1. Ductile material



- Which can be mould into wires
- D and E Far apart

2. Brittle material



- Brakes suddenly
- D and E very near

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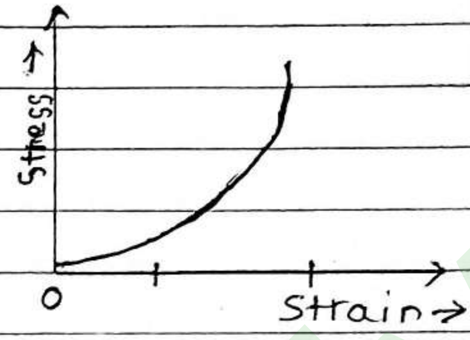
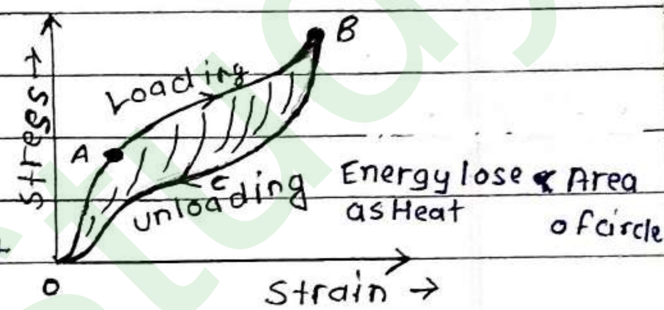
3\. Elastomers



- material which can be elastically stretched to a large volume of strain.

Elastic Hysteresis



- Do not retrace original path
- But there is no permanent strain.
- Return back completely.

- Applications of Elastic Hysteresis

→ Car tyres are made with synthetic rubbers having small area hysteresis loops so that such tyres do not excessively heated during the long journey.

→ A padding of vulcanized rubber having large area of hysteresis loop is used in shock absorbers between the vibrating system and the board. As the rubber is compressed & releasing during each vibration it dissipates a large amount of Energy.

Different types of modules

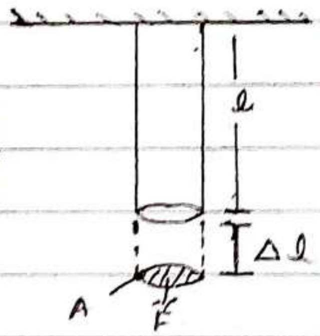
(i) Young modulus (Y) i.e modulus of elasticity of length.

(ii) Bulk modulus (K) i.e modulus of elasticity of volume.

(iii) modulus of rigidity or shet modulus (n) i.e modulus of elasticity of shape.

Young's Modulus of Elasticity (Y)

- Ratio of longitudinal stress to the longitudinal strains



$$Y = \frac{\text{longitudinal Stree}}{\text{longitudinal strain}} = \frac{\sigma}{\varepsilon}$$

$$Y = \frac{F}{A} \Rightarrow \boxed{Y = \frac{Fl}{A\Delta l}}$$

AY → stiffness

stif

Que Find the elongation of a steel bar 1m long and having cross sectional area of $1.5\,cm^2$ is subjected to a load of 1.5×10^4 N. ($Y_{stell} - 2\times10^{11}\ N/M^2$)

Solve $Y = \frac{Fl}{A\Delta l} \Rightarrow \Delta l = \frac{Fl}{AY} = \frac{1.5\times10^4 \times 1}{1.5\times10^{-4}\times 2\times10^{11}}$

$= \frac{1}{2} \times 10^{8-11} = 0.5 \times \underbrace{10^{-3}\,m}_{mm}$

$$\boxed{\Delta l = 0.5\,mm}$$

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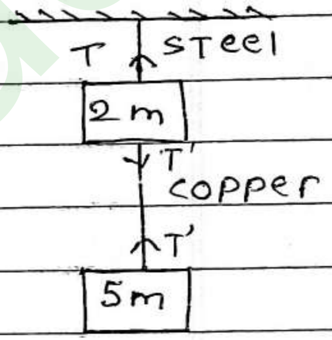
Ques IF the ratio of diameters, lengths and young's modulus of steel and copper wires shown in the figure are p, q and s respectively, then the corresponding ratio of increase in their length would be.

a. $\frac{5q}{7sp^2}$ b. $\frac{7q}{5sp^2}$ c. $\frac{2q}{5sp}$ d. $\frac{7q}{5sp}$



Soln

$\frac{d_s}{d_c} = p, \quad \frac{l_s}{l_c} = q, \quad \frac{Y_s}{Y_c} = s$

$Y = \frac{Fl}{A\Delta l}$

$\Delta l = \frac{Fl}{AY}$

$T' = F_c = 5mg$

$T = F_s = 7mg.$

$\frac{\Delta l_s}{\Delta l_c} = \cfrac{\frac{F_s l_s}{A_s Y_y}}{\frac{F_c l_c}{A_c Y_c}} = \frac{F_s l_s}{A_s Y_y} \times \frac{A_c \times Y_c}{F_c \times l_c}$

$= \left(\frac{F_s}{F_c}\right) \times \left(\frac{l_s}{l_c}\right) \times \left(\frac{A_c}{A_s}\right) \times \left(\frac{Y_c}{Y_s}\right)$

Area $= \pi\left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$

$= \left[\frac{7mg}{5mg}\right] \times q \times \left[\frac{1}{p}\right]^2 \times \left[\frac{1}{5}\right]$

$\frac{A_1}{A_2} = \left(\frac{d_1}{d_2}\right)^2 = \left(\frac{1}{p}\right)^2$

$= \frac{7q}{5} \times \frac{1}{p^2} \times \frac{1}{s}$

$\frac{\Delta l_s}{\Delta l_c} = \frac{7q}{5sp^2}$

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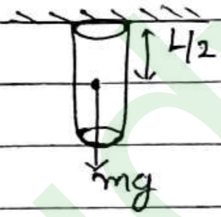
Elongation due to self weight

Que- A rod of mass m length L, Area A and longitudinal modulus Y is stre— attached to ceiling. If it is streched by its own weight. Find the elongation.



Tension varies linearly.

Sol^n

$$\Delta l = \frac{Fl}{AY}$$

$$= \frac{mg \times \frac{l}{2}}{AY}$$

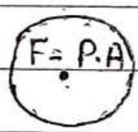
$$\boxed{\Delta l = \frac{mgl}{2AY}}$$

Interms of density $\boxed{\Delta l = \frac{\rho g l}{2\gamma}}$

Bulk Modulus of Elasticity (β) (K)

The ratio of hydraulic / volumetric stress to the corresponding hydraulic strain is called bulk modulus.

$$B = \frac{\text{Volumetric stress}}{\text{Vol. strain}}$$



F = PA

$$= \frac{\frac{F}{A}}{\frac{\Delta V}{V}} = \frac{P}{\frac{-\Delta V}{\Delta}}$$

Net compressing force

$F_{net} = PA - P_o A$

$= A(P - P_o)$

$\text{Stress} = \frac{F_{net}}{A} = (P - P_o) = \Delta P$

$$\boxed{\beta = \frac{\Delta P}{\frac{\Delta V}{V}}}$$

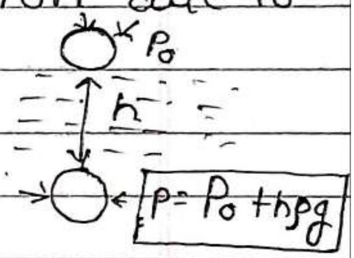
unit = Nm^{-2} or Pascal.

→ Compressibility

Reciprocal of the bulk modulus of the material of a body is called compressibility of that material.

$$\left[C = \frac{1}{\beta}\right]$$

Que To what depth below the surface of sea should a rubber ball be taken as to decrease its volume by 0.1% (Take density of sea water = 1000 $kg\,m^{-3}$, Bulk modulus of rubber = $9 \times 10^8\ Nm^{-2}$, acceleration due to gravity = 10 m/s^2)



Solⁿ

$$\frac{\Delta V}{V}\% = 0.1\% = \frac{0.1}{100} = 10^{-3}$$

$$\beta = \frac{\Delta P}{\frac{\Delta V}{V}} \Rightarrow 9 \times 10^8 = \frac{h\rho g}{10^{-3}} \qquad (\Delta P = h\rho g)$$

$$9 \times 10^8 = \frac{h \times 10^3 \times 10}{10^{-3}}$$

$$h = \frac{9 \times 10^6}{10^4} = 90m$$

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Ques The approximate depth of an ocean is 2700m. The compressibility of water is $45.4 \times 10^{-11}\ Pa^{-1}$ and density of water is $10^3\ kg/m^3$. what fractional compression of water will be obtained at the bottom of the ocean.

Solⁿ

$h = 2700\ m$

$C = 45.4 \times 10^{-11}\ Pa^{-1}$

$\rho = 10^3\ kg/m^3$

$\frac{\Delta V}{V} = ?$

$$B = \frac{\Delta P}{\frac{\Delta V}{V}} \Rightarrow \frac{1}{C} = \frac{h\rho g}{\left(\frac{\Delta V}{V}\right)} \Rightarrow \frac{\Delta V}{V} = h\rho g \times C$$

$$= 2700 \times 10^3 \times 10 \times 45.4 \times 10^{-11}$$

$$= 27 \times 454 \times 10^{-6}$$

$$= 12258 \times 10^{-6}$$

$$= 1.2258 \times 10^{-2}$$

Relation between Density (ρ), Pressure (P) & Bulk modulus (B)

initia density = ρ

After compression, density = ρ'

$$\boxed{\rho' = \rho\left(1 + \frac{\Delta P}{B}\right)}$$

Ques The bulk modulus of water is $2.1 \times 10^9 \, Nm^{-2}$. The pressure required to increase the density of water by 0.1% is?

Sol^n

$$B = 2.1 \times 10^9 \, Nm^{-2}$$

$$\frac{\Delta \rho}{\rho} \% = 0.1\% = \frac{0.1}{100} = \frac{1}{100} = 10^{-3}$$

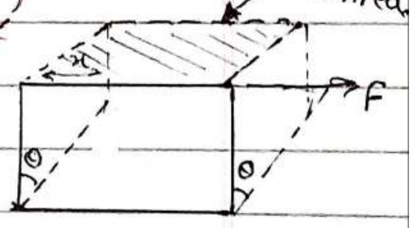
$$\frac{\Delta \rho}{\rho} = \frac{\rho' - \rho}{\rho}$$

$$\frac{\Delta \rho}{\cancel{\rho}} = \frac{\rho\left(1 + \frac{\Delta P}{B}\right) - \rho}{\cancel{\rho}\ \rho}$$

$$10^{-3} = \cancel{1} + \frac{\Delta P}{\beta} - \cancel{1}$$

$$10^{-3} = \frac{\Delta P}{2.1 \times 10^9} \Rightarrow \boxed{\Delta P = 2.1 \times 10^6}$$

Shear Modulus of Rigidity (η)



$$\eta = \frac{\text{Shearing stress}}{\text{Shearing strain}}$$

$$= \frac{\frac{F_t}{A}}{\frac{x}{l}} = \frac{\frac{F_t}{A}}{\tan\theta} = \frac{F_t}{A\tan\theta}$$

$$\boxed{\eta \text{ or } G = \frac{F}{A\tan\theta}}$$

If $x <<< L$

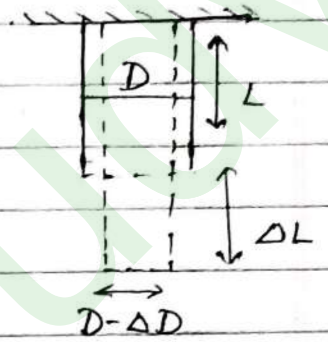
$\tan\theta = \theta$

$$\left(\eta = \frac{F}{A\theta}\right)$$

→ A solid will have all three modulus Y, B & η

→ But in case of liquid or gases only B can be defined.

Poison's Ration (σ_p)



$$\sigma_p = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{\frac{\Delta D}{D}}{\frac{\Delta L}{L}}$$

$$\left\{ \sigma_p = \frac{\Delta D L}{D \Delta L} \right\}$$

Range of σ_p ⇒

$-1 \le \sigma_p \le 0.5$

Particle, $\sigma_p = 0$ e.g. cork

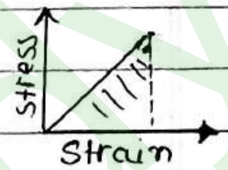
Potential Energy stored in Streched wire

$$\left[U = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}\right]$$

$Y = \frac{\text{stress}}{\text{strain}}$

OR

$$\left[U = \frac{1}{2} \times Y \times (\text{strain})^2 \times \text{volume}\right]$$



U = area under curve

OR

$$\left[U = \frac{1}{2} \times \left(\frac{\text{stress}}{Y}\right)^2 \times \text{Volume}\right]$$

Y = Slope of stree-strain curve.

OR

$$\left[U = \frac{1}{2} \times \underset{(F)}{\text{load}} \times \underset{(\Delta l)}{\text{Elongation}}\right]$$

workdone = Elastic P.E

- when wire is streched, work is done against restoring Forces acting between particles of wire.

- Stored work done is its elastic potential energy.

$$U = \frac{1}{2} \times \sigma \times \epsilon \times V \Rightarrow \frac{U}{V} = \frac{1}{2}\sigma\epsilon$$

$$\left[u = \frac{1}{2}\sigma\epsilon\right]$$

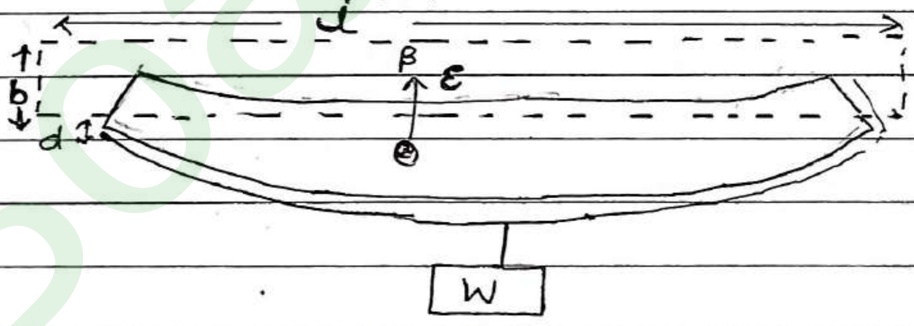
Application of Elastic behaviour of materials

1. Metallic part of a machinery is never subjected to a stress beyond the elastic limit because beyond it, it will permanently deform that metallic part.

2. Thickness of metallic ropes used in cranes to lift heavy loads is decided from the knowledge of the elastic limit of a material and safety factor.

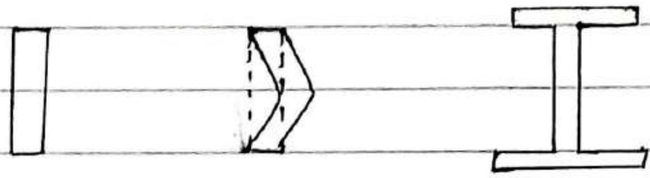
3. In designing a bridge such that it does not bend too much or break under the load of traffic, the force of wind and under it own weight

$$\delta = \frac{wl^3}{4Ybd^3}$$



→ Depression can be decreased more effectively by increasing the depth rather than bredth.

- But deep bar has a tendency to bend under weight of a moving traffic. Bending is called Buckling. Hence, better choice is to have I-shaped.



Bulking.

4. The maximum height of moutain on earth depends upon shear modulus of rock.

The material at the base experiences this force per unit area in the verticle direction, but sides of moutain are free.

5. A hollow shaft is stronger than a solid shaft made of equal quantity of some material. As the torque required to twist the hollow cylider is greater than torque of solid.